

IDW and ILDW (Short Pseudocode)

Coordinate convention (used in both algorithms). All points are in local patch coordinates, in meters, with origin at the NW (top-left) corner. x increases to the east, y increases to the south. Thus $(x0_k, y0_k)$ and $(x1_k, y1_k)$ are link endpoints in this frame, and q_t is the center of pixel t in the same frame.

IDW

Algorithm 1 Discretized Midpoint IDW

Require: links $(x0_k, y0_k, x1_k, y1_k, A_k, f_k, \text{pol}_k)$, radius $r_{\max}$ (default: 15000 m), power p (default: 2), default d_0 (default: 0)

- 1: **for** each link k **do**
- 2: midpoint $m_k \leftarrow ((x0_k + x1_k)/2, (y0_k + y1_k)/2)$
- 3: $\ell_k^{(m)} \leftarrow \|(x1_k, y1_k) - (x0_k, y0_k)\|$ ▷ computed in meters
- 4: $L_k \leftarrow \ell_k^{(m)}/1000$ ▷ converted and stored in km
- 5: $(\kappa_k, \alpha_k) \leftarrow \text{ITU838}(f_k, \text{pol}_k)$
- 6: $R_k \leftarrow \left(\frac{A_k/L_k}{\kappa_k} \right)^{1/\alpha_k}$ ▷ link value, located at midpoint m_k
- 7: **end for**
- 8: **for** each pixel t with center q_t **do**
- 9: **if** one or more link midpoints lie inside pixel t **then**
- 10: $\hat{R}_t \leftarrow$ mean of R_k of those links
- 11: **else**
- 12: $N_t \leftarrow \{k : \|m_k - q_t\| \leq r_{\max}\}$
- 13: **if** $N_t = \emptyset$ **then**
- 14: $\hat{R}_t \leftarrow d_0$
- 15: **else**
- 16: $\tilde{w}_k \leftarrow \|m_k - q_t\|^{-p}$ for $k \in N_t$
- 17: $w_k \leftarrow \frac{\tilde{w}_k}{\sum_{j \in N_t} \tilde{w}_j}$ for $k \in N_t$
- 18: $\hat{R}_t \leftarrow \sum_{k \in N_t} w_k R_k$
- 19: **end if**
- 20: **end if**
- 21: **end for**
- 22: Return field $\hat{R}$

ILDW

Algorithm 2 Segment-Distance ILDW

Require: links $(a_k, b_k, A_k, f_k, \text{pol}_k)$, radius $r_{\max}$ (default: 15000 m), power p (default: 2), default d_0 (default: 0)

- 1: **for** each link k **do**
- 2: midpoint $m_k \leftarrow (a_k + b_k)/2$
- 3: $\ell_k^{(m)} \leftarrow \|b_k - a_k\|$ ▷ computed in meters
- 4: $L_k \leftarrow \ell_k^{(m)}/1000$ ▷ converted and stored in km
- 5: $(\kappa_k, \alpha_k) \leftarrow \text{ITU838}(f_k, \text{pol}_k)$
- 6: $R_k \leftarrow \left(\frac{A_k/L_k}{\kappa_k} \right)^{1/\alpha_k}$ ▷ link value; used on segment $[a_k, b_k]$
- 7: **end for**
- 8: **for** each pixel t with center q_t **do**
- 9: **if** one or more links cross pixel t **then**
- 10: $\hat{R}_t \leftarrow$ mean of R_k of those links
- 11: **else**
- 12: $N_t \leftarrow \{k : \text{dist}(q_t, [a_k, b_k]) \leq r_{\max}\}$ ▷ dist is point-to-segment distance
- 13: **if** $N_t = \emptyset$ **then**
- 14: $\hat{R}_t \leftarrow d_0$
- 15: **else**
- 16: $\tilde{w}_k \leftarrow \text{dist}(q_t, [a_k, b_k])^{-p}$ for $k \in N_t$
- 17: $w_k \leftarrow \frac{\tilde{w}_k}{\sum_{j \in N_t} \tilde{w}_j}$ for $k \in N_t$
- 18: $\hat{R}_t \leftarrow \sum_{k \in N_t} w_k R_k$
- 19: **end if**
- 20: **end if**
- 21: **end for**
- 22: Return field $\hat{R}$
